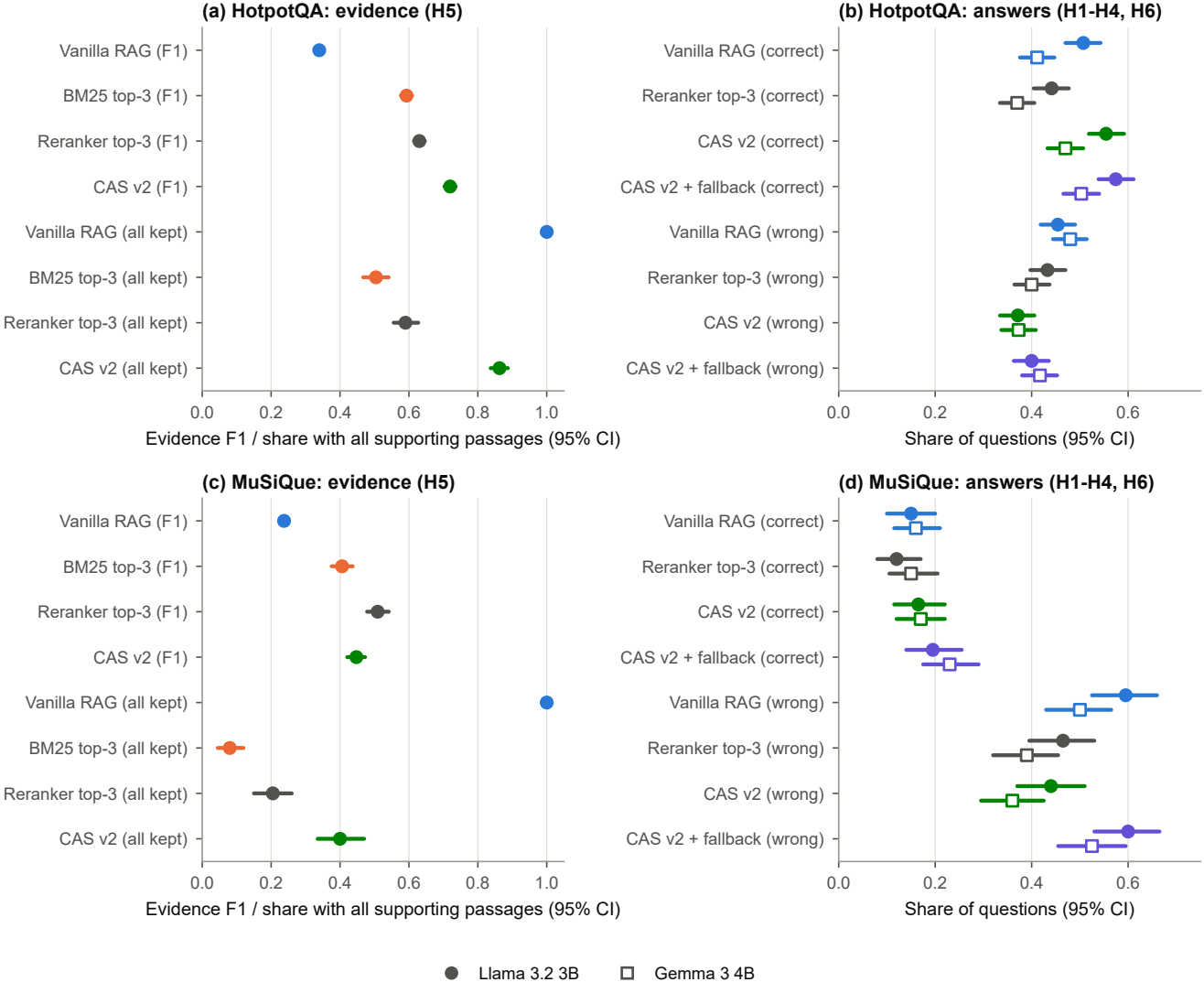

(a) HotpotQA: evidence (H5)
Vanilla RAG (F1)
BM25 top-3 (F1)
Reranker top-3 (F1)
CAS v2 (F1)
Vanilla RAG (all kept)
BM25 top-3 (all kept)
Reranker top-3 (all kept)
CAS v2 (all kept)
0.0
0.2
0.4
0.6
0.8
1.0
Evidence F1 / share with all supporting passages (95% CI)
(b) HotpotQA: answers (H1-H4, H6)
Vanilla RAG (correct)
Reranker top-3 (correct)
CAS v2 (correct)
CAS v2 + fallback (correct)
Vanilla RAG (wrong)
Reranker top-3 (wrong)
CAS v2 (wrong)
CAS v2 + fallback (wrong)
0.0
0.2
0.4
0.6
Share of questions (95% CI)
(c) MuSiQue: evidence (H5)
Vanilla RAG (F1)
BM25 top-3 (F1)
Reranker top-3 (F1)
CAS v2 (F1)
Vanilla RAG (all kept)
BM25 top-3 (all kept)
Reranker top-3 (all kept)
CAS v2 (all kept)
0.0
0.2
0.4
0.6
0.8
1.0
Evidence F1 / share with all supporting passages (95% CI)
(d) MuSiQue: answers (H1-H4, H6)
Vanilla RAG (correct)
Reranker top-3 (correct)
CAS v2 (correct)
CAS v2 + fallback (correct)
Vanilla RAG (wrong)
Reranker top-3 (wrong)
CAS v2 (wrong)
CAS v2 + fallback (wrong)
0.0
0.2
0.4
0.6
Share of questions (95% CI)
Llama 3.2 3B
Gemma 3 4B